

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech.

Branch : Electrical Engineering & Allied Semester :IV

Subject Code & Name: BTEEPE405C Advanced Renewable Energy Sources

Max Marks: 60

Date: 31/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following convert's energy from the combustion of fuel directly to the electrical energy?	Understand	1
	a. Ni-Cd cell b. Dynamo c. Fuel cell d. Electrolytic cell		
2	Chemical energy is converted to _____ energy by a fuel cell.	Remember	1
	a. solar b. electrical c. potential d. mechanical		
3	An electro-mechanical energy conversion device is one which converts _____	Remember	1
	a. Electrical energy to mechanical energy only b. Mechanical energy to electrical energy only c. Electrical to mechanical and mechanical to electrical d. None of the mentioned		
4	Which of the following components of a wind turbine system is arranged in the order in which energy is being converted?	Understand	1
	a. Blades – rotor – electric generator – shaft b. Blades – rotor – shaft – electric generator c. Shaft – blades – rotor – electric generator d. Electric generator – blades – rotor – shaft		

5	Which of the following is an advantage of a horizontal axis wind turbine?				Understand	1
	a. Blades are to the side of turbine's center of gravity	b. Blades are parallel to the vertical axis passing through the turbine's center of gravity	c. Blades are parallel to the horizontal axis passing through the turbine's center of gravity	d. Blades are positioned perpendicular to the plane of the ground		
6	13. What are the types of VAWT?				Remember	1
	a. Lift, horizontal	b. Horizontal, vertical	c. Drag, lift	d. Urban wind turbines		
7	The current produce by the solar cell can be given by _____				Remember	1
	a. $IL - ID + ISh$	b. $IL + ID - ISh$	c. $IL + ID + ISh$	d. $IL - ID - ISh$		
8	In solar power plant, the output terminals of solar photovoltaic array can be directly connected to _____				Understand	1
	a. AC to DC Converter	b. DC Bus Bar	c. AC Bus Bar	d. AC Load		
9	A Solar thermal collector				Remember	1
	a. collects solar energy and reflects it back	b. Absorb the solar radiation	c. collects and convert solar energy into electrical energy	d. collects and converts solar energy into thermal energy and delivers it to the next stage of system		
10	Which of the following is a substrate for biogas production?				Understand	1
	a. Municipal & residential waste	b. E-waste	c. Metallic waste	d. Gaseous effluents		
11	Which of the following is used to produce biogas from biomass?				Understand	1
	a. Anaerobic	b. Aerobic	c.	d. Pyrolysis		

	treatment	treatment	Fermentation			
12	The function of flywheel is				Remember	1
	a.to convert reciprocating motion of the piston into rotary motion	b. to convert rotary motion of crankshaft into to and fro motion of the valve rod	c. to prevent fluctuation of speed	d. to keep the engine speed uniform at all load condition		
Q. 2	Solve the following.					12
A)	Discuss renewable and conventional forms of energy. Highlight their merits and demerits?				Understand	6
B)	Explain the construction and working principle of Proton exchange membrane fuel cell?				Remember	6
Q.3	Solve the following.					12
A)	Explain the factors considered for appropriate site selection of wind farm?				Understand	6
B)	Draw Wind Power generation curve and explain the terms cut-in-speed, rated speed and cut-out speed.				Remember	6
Q. 4	Solve Any Two of the following.					12
A)	Draw and explain current voltage characteristics of solar PV cell. Also define fill factor?				Remember	6
B)	Draw the equivalent circuit of solar cell and deduce the relation $V_{oc} = AKT/q \cdot \ln[(I_{sc}/I_o) + 1]$ where the symbols have their usual significance				Remember	6
C)	Discuss the standalone type PV system.				Understand	6
Q.5	Solve Any Two of the following.					12
A)	With the help of neat sketch, explain the classification of biogas plant?				Understand	6
B)	With the neat diagram discuss the biogas gasifiers?				Understand	6
C)	Explain in detail the small scale pyrolysis Unit with the help of neat diagram?				Remember	6

Q. 6	Solve Any Two of the following.		12
A)	Explain super conducting Magnetic storage system with a neat diagram?	Understand	6
B)	Draw schematic diagram of lead acid cell and explain its working in detail?	Remember	6
C)	Explain in detail the thermal energy storage?	Remember	6
*** End ***			